SOC THEORETICAL TASK REPORT

Name: Atharva Balwadkar
Organization: CyArt
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Introduction

Security Operations Centers (SOCs) play a critical role in monitoring, detecting, analyzing, and responding to cybersecurity threats in real time. SOC analysts continuously monitor alerts generated from various security tools such as SIEM, IDS, and endpoint protection systems.

To effectively manage security incidents, analysts must understand how to prioritize alerts, classify incidents, and respond using a structured incident response process. Proper alert prioritization ensures that the most critical threats are addressed first, reducing potential damage to organizational systems and data.

This report discusses key SOC concepts including alert priority levels, incident classification frameworks, and the basic incident response lifecycle used in modern cybersecurity operations.

1. Alert Priority Levels

Alert priority levels help SOC analysts determine how severe a security alert is and how quickly it must be addressed. Since security monitoring systems generate thousands of alerts daily, prioritization helps analysts focus on the most critical threats first.

Alerts are typically categorized into four severity levels: Critical, High, Medium, and Low.

1.1 Priority Definitions

Critical

Critical alerts represent the highest severity incidents and require immediate response. These alerts often indicate an active compromise or severe security threat.

Examples include:

- Ransomware encrypting files
- Domain controller compromise
- Data exfiltration
- Active exploitation of a vulnerability

Such alerts can lead to major business disruption or data breaches.

High

High priority alerts indicate serious security threats that require immediate investigation but may not yet have caused full compromise.

Examples include:

- Unauthorized administrator login
- Exploitation attempts against servers
- Suspicious privilege escalation activity

These alerts could lead to severe incidents if not addressed quickly.

Medium

Medium priority alerts represent suspicious activity that may indicate an attack but requires further investigation.
Examples include:

- Multiple failed login attempts
- Unusual network traffic patterns
- Possible brute-force attacks

While these alerts may not immediately affect systems, they can signal potential threats.

Low

Low priority alerts are informational alerts with minimal security risk.
Examples include:

- Port scanning activity
- Minor configuration changes
- Normal system events

These alerts are usually monitored but do not require immediate action.

1.2 Alert Prioritization Criteria

SOC analysts prioritize alerts using several important factors.

- **Asset Criticality**

Assets such as production servers, databases, or domain controllers are more critical than test machines. Alerts affecting critical assets are assigned higher priority.

- **Exploit Likelihood**

If a vulnerability has publicly available exploit code, the risk of attack increases significantly. For example, the Log4Shell vulnerability (CVE-2021-44228) had a CVSS score of 9.8, which classifies it as a critical vulnerability.

- **Business Impact**

Incidents that can cause financial loss, operational disruption, or reputational damage are treated with higher urgency.

1.3 CVSS Scoring System

The **Common Vulnerability Scoring System (CVSS)** is an open industry standard for assessing the severity of computer system security vulnerabilities, producing a numerical score from 0.0 to 10.0, where 10.0 is the most critical. It provides a standardized method to prioritize remediation based on base, temporal, and environmental metrics.

CVSS Score	Severity Level
0.0 – 3.9	Low
4.0 – 6.9	Medium
7.0 – 8.9	High
9.0 – 10.0	Critical

SOC teams use CVSS scores to prioritize vulnerabilities and security alerts effectively.

2. Incident Classification

Incident classification is the process of categorizing security incidents based on their type, impact, and attack method. Proper classification helps security teams quickly understand the nature of an incident and respond appropriately.
Standardized frameworks such as MITRE ATT&CK, ENISA Incident Taxonomy, and VERIS are widely used to classify cybersecurity incidents.

2.1 Common Incident Categories

Malware

Malware refers to malicious software designed to damage systems or gain unauthorized access.
Examples include:

- Ransomware
- Trojans
- Spyware
- Worms

Malware infections can lead to data theft, system compromise, or operational disruption.

Phishing

Phishing attacks involve fraudulent emails or websites designed to trick users into revealing sensitive information such as login credentials or financial data.
These attacks are one of the most common entry points for cyber attackers.

Distributed Denial of Service (DDoS)

DDoS attacks attempt to overwhelm servers or networks with massive traffic, making services unavailable to legitimate users.
Organizations often experience service downtime during such attacks.

Insider Threat

Insider threats originate from employees, contractors, or trusted individuals within an organization who misuse their access privileges.
Example:
An employee exporting confidential data without authorization.

Data Exfiltration

Data exfiltration refers to the unauthorized transfer of sensitive data outside the organization. This may involve customer data, intellectual property, or confidential business information.

2.2 MITRE ATT&CK Framework

The MITRE ATT&CK framework is widely used to classify and analyze attacker behavior based on tactics and techniques.

Technique ID	Description
T1566	Phishing
T1190	Exploit Public Facing Application
T1078	Valid Accounts
T1046	Network Service Scanning

Using this framework allows SOC analysts to map alerts to known attack patterns.

2.3 Contextual Metadata

To improve incident investigation, SOC analysts enrich alerts with contextual metadata such as:

- Source IP address
- Target system
- Timestamp
- File hashes
- Indicators of Compromise (IOCs)

This additional information helps analysts understand the scope and impact of security incidents.

3. Basic Incident Response

Incident response is a structured process used by organizations to detect, analyze, and respond to cybersecurity incidents. A well-defined incident response strategy helps minimize damage, recover systems quickly, and prevent future attacks.
Most organizations follow the NIST Incident Response Lifecycle, which consists of six phases.

3.1 Incident Response Lifecycle

1. Preparation

Preparation involves developing policies, procedures, and tools necessary to respond to security incidents.
This includes:

- Incident response playbooks
- Security monitoring tools
- Training SOC analysts

Preparation ensures the organization is ready to handle potential cyber threats.

2. Identification

During this phase, SOC analysts detect potential security incidents through alerts, logs, or monitoring systems.
The analyst verifies whether the alert represents a real security incident or a false positive.

3. Containment

Once an incident is confirmed, the affected systems are isolated to prevent the attack from spreading.
Examples include:

- Disconnecting compromised machines from the network
- Blocking malicious IP addresses
- Disabling compromised accounts

4. Eradication

In the eradication phase, the root cause of the attack is removed from the environment.
Examples include:

- Removing malware
- Deleting malicious files
- Patching vulnerabilities

5. Recovery

Recovery involves restoring systems to normal operations.
Actions may include:

- Restoring systems from backups
- Monitoring systems for unusual activity
- Ensuring vulnerabilities are fixed

6. Lessons Learned

After the incident is resolved, the security team conducts a post-incident review to analyze what happened and how to improve the response process.
This phase helps organizations strengthen their security posture and prevent future incidents.

Conclusion

In modern cybersecurity environments, Security Operations Centers play a crucial role in protecting organizational systems and data from cyber threats. Effective SOC operations depend on the ability to properly prioritize alerts, accurately classify incidents, and respond quickly using a structured incident response process.
Understanding alert priority levels helps SOC analysts identify which threats require immediate attention and ensures that critical incidents are handled without delay. Incident classification frameworks such as MITRE ATT&CK provide a standardized way to categorize and analyze different types of cyberattacks, improving investigation efficiency.
Additionally, the incident response lifecycle provides a systematic approach for handling security incidents, from detection and containment to recovery and lessons learned. By following these structured processes, organizations can minimize damage, restore normal operations quickly, and strengthen their overall cybersecurity posture.